(1) Consider the uncoupled system

$$x' = 3x \quad y' = -y$$

a) Solve the system explicitly for the initial conditions $(x(0), y(0)) = (x_0, y_0)$

$$\frac{dx}{dt} = 3x \Rightarrow dx = 3x \cdot dt$$

$$\Rightarrow 1/x \, dx = 3 \, dt \Rightarrow \ln|x| = 3t + C_1$$

$$x = C_1 e^{3t}$$

$$x(0) = x_0 = C_1 e^0 = C_1 = x_0$$

$$\frac{dy}{dt} = -y \Rightarrow \int dy \, 1/y = -t + C_2$$

$$\ln|y| = -t + C_2 \Rightarrow y = e^{-t + C_2}$$

$$= e^{C_2} e^{-t} = C_2 e^{-t}$$

$$y(0) = y_0 = C_2 e^0 = C_2 = y_0$$

- For $(x(0), y(0)) = (x_0, y_0)$ we have

$$x = x_0 e^{3t}$$

$$y = y_0 e^{-t}$$

b) $x' = 3x \quad y' = -y$

Find the equilibrium points

$x' = 3x = 0$ when $x = 0$

$y' = -y = 0$ when $y = 0$

- There are no others

c) Describe the behavior as $t \to \infty$ for

i) initial conditions on the x-axis

$x = x_0 e^{3t}$ as $t \to \infty$, $x \to \infty$

ii) $y = y_0 e^{-t}$ as $t \to \infty$, $y \to 0$

iii) initial conditions not on either axis

$x = x_0 e^{3t}$

$y = y_0 e^{-t}$

By cases:

if $x_0 < 0$ & $y_0 < 0$

$x \to -\infty$ & $y \to 0$

if $x_0 > 0$ & $y_0 < 0$

$x \to \infty$ & $y \to 0$

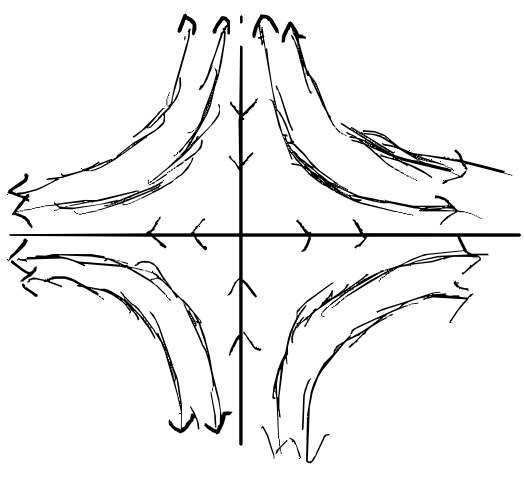
if $x_0 < 0$ & $y_0 > 0$

$x \to -\infty$ & $y \to 0$

if $x_0 > 0$ & $y_0 > 0$

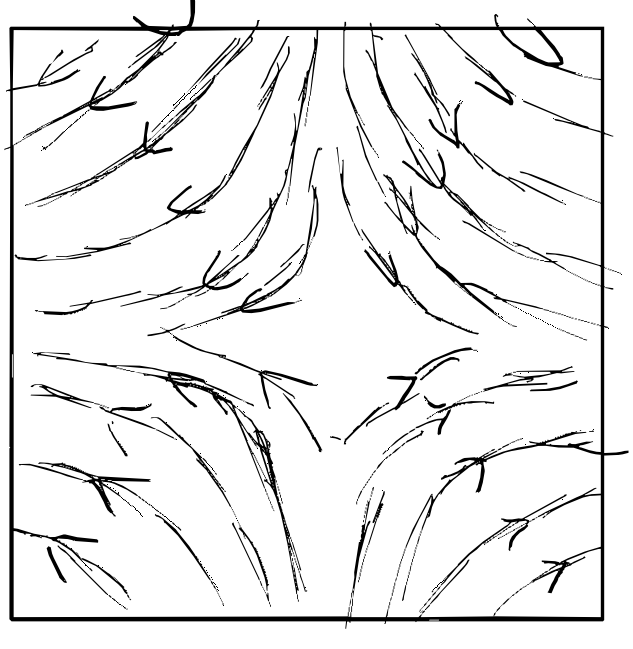
$x \to \infty$ & $y \to 0$

d) Identify the equilibrium point.



- Totally a saddle

e)



- just reusing the code from HW #1

(2) Consider the system

$x' = -5x$

$y' = -y$

a) Solve For $(x(0), y(0)) = (x_0, y_0)$

$\frac{dx}{dt} = -5x \Rightarrow \ln|x| = -5t + C_1$

$\Rightarrow x = e^{-5t + C_1} = C_1 e^{-5t}$

$\Rightarrow x = x_0 e^{-5t}$

$\frac{dy}{dt} = -y \Rightarrow \ln|y| = -t + C_2$

$\Rightarrow y = y_0 e^{-t}$

- For $(x(0), y(0)) = (x_0, y_0)$

$x = x_0 e^{-5t}$

$y = y_0 e^{-t}$

(b) Determine which variable decays Faster as $t \to \infty$

$\lim_{t \to \infty} \frac{x_0 e^{-5t}}{y_0 e^{-t}} = e^{-5t+t} = e^{-4t}$

- We're decaying Faster in the x direction

(c) Show that For any solution w/ $x_0 \neq 0$ $\lim_{t \to \infty} y(t)/x(t) = \infty$

$\lim_{t \to \infty} \frac{y(t)}{x(t)} = \lim_{t \to \infty} \frac{y_0 e^{-t}}{x_0 e^{-5t}}$

$= \lim_{t \to \infty} \frac{y_0}{x_0} e^{4t}$

By cases

$x<0, y<0$ $\lim_{t \to \infty} \frac{y_0}{x_0} e^{4t} = \infty$

$x<0, y>0$ $\lim_{t \to \infty} \frac{y_0}{x_0} e^{4t} = -\infty$

$x>0, y<0$ $\lim_{t \to \infty} \frac{y_0}{x_0} e^{4t} = -\infty$

$x<0, y<0$ $\lim_{t \to \infty} \frac{y_0}{x_0} e^{4t} = \infty$

Diverges slightly From whats stated, but, we're headed either vertical up or vertical down For large t

(d) Conclude that we converge to the y axis For large t

- We know From above, $x' = -5x$ and $\lim_{t \to \infty} \frac{y(t)}{x(t)} = \pm\infty$ that we're decreasing exponentially in x while becoming asymptotic to y. That should be sufficent

(e) Demonstrate similar end behavior For $t \to -\infty$.

$x<0, y<0$ $\lim_{t \to -\infty} \frac{y_0}{x_0} e^{4t} = 0$

$x<0, y>0$ $\lim_{t \to -\infty} \frac{y_0}{x_0} e^{4t} = 0$

$x>0, y<0$ $\lim_{t \to -\infty} \frac{y_0}{x_0} e^{4t} = 0$

$x<0, y<0$ $\lim_{t \to -\infty} \frac{y_0}{x_0} e^{4t} = 0$

We can do likewise as before.

Examine $y = y_0 e^{-t}$

$\lim_{t \to -\infty} y = \lim_{t \to -\infty} y_0 e^{-t} = \lim_{t \to \infty} y_0 e^{t} = \infty$

The behavoir Flips. We start approaching the x-axis.

(3) Show that for a linear & homogeneous system of ODE w/ $a,b,c,d \neq 0$ where x' & y' are not scalar multiples (linearly dependent)

$$x' = ax + by \quad \& \quad y' = cx + dy$$

only have an equilibrium point at $(0,0)$

- We're given that x' & y' are not scalar multiples, that gives us that $\nexists s \in \mathbb{R}$ such that $sx' = y'$. As, if expanded this implies

$$\nexists s \in \mathbb{R} \text{ s.t. } \begin{aligned} sx' &= s \cdot ax + s \cdot by \\ &= y' = cx + dy \end{aligned}$$

and that

$$\nexists s \in \mathbb{R} \text{ s.t. } s \cdot ax = cx \text{ and } s \cdot by = dy$$

With a little rearranging

$$s \cdot ax = cx \qquad s \cdot by = dy$$
$$s \cdot a = c \qquad s \cdot b = d$$

so s would have to fulfill

$$s = c/a \quad \& \quad s = d/b$$
$$c/a = s = d/b$$
$$c/a = d/b$$
$$ad = bc$$
$$ad - bc = 0$$

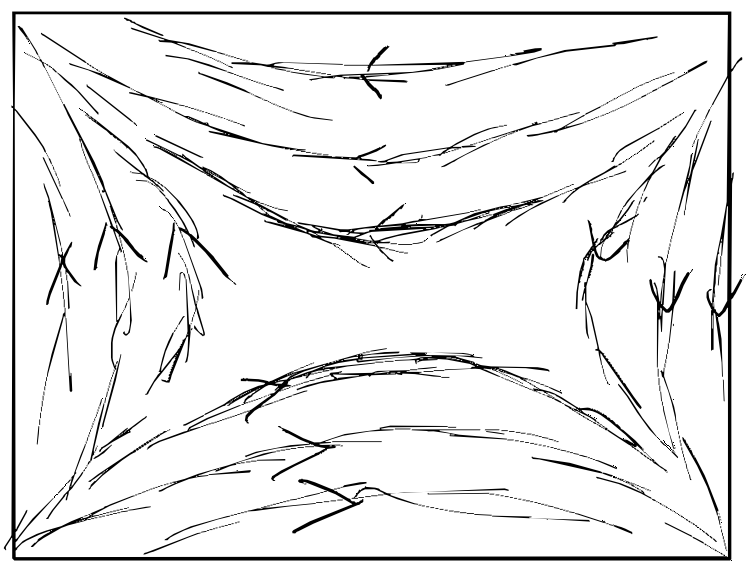
But, as we know this can't be the case, as $\nexists s \in \mathbb{R}$, for a 2×2 where $\det(A) = ad - bc$ has to be nonzero, thus invertible, thus the only only solution is

$$(0,0)$$

(4) Given $x' = -y$ & $y' = -x$

a) Plot using technology, find a few curves

- same code as HW #1



- copy/paste on remarkable is not a thing

b) We want to show that these are of the form $x^2 - y^2 = C$, where C is constant.

i) Convert to $x\frac{dx}{dt} - y\frac{dy}{dt} = 0$

$$x' = -y \qquad y' = -x$$

$$\frac{dx}{dt} = -y \qquad \frac{dy}{dt} = -x$$

$$x\frac{dx}{dt} = -y \cdot x \qquad y\frac{dy}{dt} = -x \cdot y$$

$$x\frac{dx}{dt} - y\frac{dy}{dt} = -xy + xy = 0$$

$$x\frac{dx}{dt} - y\frac{dy}{dt} = 0$$

ii) Use the chain rule to show that

$$\tfrac{1}{2}\, d/dt(x^2) = x\, dx/dt$$

$$\begin{aligned}\tfrac{1}{2}\, d/dt(x^2) &= \tfrac{1}{2}\, d/dt(x(t)^2) \\ &= \tfrac{1}{2} \cdot 2 \cdot x(t)\, d/dt(x(t)) \\ &= x(t)\, dx/dt \\ &= x\, dx/dt\end{aligned}$$

iii) Substitute this back for both sides

$$\tfrac{1}{2}\, d/dt(x^2) = x\, dx/dt \qquad \tfrac{1}{2}\, d/dt(y^2) = y\, dy/dt$$

$$x\frac{dx}{dt} - y\frac{dy}{dt} = 0$$

$$\tfrac{1}{2}\, d/dt\, x^2 - \tfrac{1}{2}\, d/dt\, y^2 = 0$$

iv) Use FTC to get $x^2 - y^2 = C$

$$\frac{1}{2}\int d/dt(x^2)\, dt - \frac{1}{2}\int d/dt(y^2)\, dt = 0$$

$$\tfrac{1}{2}x^2 - \tfrac{1}{2}y^2 = C$$

$$x^2 - y^2 = C$$

(5) Consider

$$x' = -y \quad \& \quad y' = -x$$

We want to decouple and solve this automatically

a) introduce new variables

$$u = x+y \quad \& \quad v = x-y$$

$$u+v = (x+y)+(x-y) = 2x \qquad u-v = (x+y)-(x-y) = 2y$$

$$\frac{u+v}{2} = x \qquad \frac{u-v}{2} = y$$

b) $u = x+y$

$u' = x' + y'$

$= -y - x$

$= -u$

$u' = -u$

$v = x-y$

$v' = x' - y'$

$= -y + x$

$= v$

$v' = v$

c) Solve for $u(t)$ & $v(t)$ w/ arbitrary I.C. (u_0, v_0)

$$du/dt = -u \qquad dv/dt = v$$

$$du = -u \cdot dt \qquad dv = v \cdot dt$$

$$\frac{1}{u} du = -1 \cdot dt \qquad \frac{1}{v} dv = 1 \cdot dt$$

$$\ln|u| = -t + C_1 \qquad \ln|v| = t + C_2$$

$$u = e^{-t} e^{C_1} \qquad v = e^{t} e^{C_2}$$

let $C_1 = e^{C_1}$ & $C_2 = e^{C_2}$

$$u = C_1 e^{-t} \quad \& \quad v = C_2 e^{t}$$

- For initial conditions

$$u_0 = u(0) = C_1 e^0 = C_1 \qquad v_0 = v(0) = C_2 e^0 = C_2$$

$$u(t) = u_0 e^{-t} \qquad v(t) = v_0 e^{t}$$

d) $u = x+y$ & $v = x-y$

$$\frac{u+v}{2} = x \qquad \frac{u-v}{2} = y$$

Now we want above back in terms of x & y

$$x = \frac{u(t)+v(t)}{2} \qquad y = \frac{u(t)-v(t)}{2}$$

$$x = \frac{u_0 e^{-t} + v_0 e^{t}}{2} \qquad y = \frac{u_0 e^{-t} - v_0 e^{t}}{2}$$

$$x = \frac{1}{2} u_0 e^{-t} + \frac{1}{2} v_0 e^{t} \qquad y = \frac{1}{2} u_0 e^{-t} - \frac{1}{2} v_0 e^{t}$$

- Looks like hyperbolic trig

$$\sinh(x) = \frac{e^{x} - e^{-x}}{2} \qquad \cosh(x) = \frac{e^{x} + e^{-x}}{2}$$

- Above will suffice though